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Quantum Circuits for the Black-Scholes equations via Schrödingerisation

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Abstract

In this paper, we construct quantum circuits for the Black-Scholes equations, a cornerstone of financial modeling, based on a quantum algorithm that overcome the cure of high dimensionality. Our approach leverages the Schrödingerisation technique, which converts linear partial and ordinary differential equations with non-unitary dynamics into a system evolved by unitary dynamics. This is achieved through a warped phase transformation that lifts the problem into a higher-dimensional space, enabling the simulation of the Black-Scholes equation on a quantum computer. We will conduct a thorough complexity analysis to highlight the quantum advantages of our approach compared to existing algorithms. The effectiveness of our quantum circuit is substantiated through extensive numerical experiments.

Keywords: Black-Scholes equations, Schrödingerisation, Quantum circuits

1 Introduction

The Black-Scholes equation, first introduced in 1973, is an important mathematical model in the field of derivatives pricing, which lays a theoretical foundation for derivatives pricing and promotes further research on mathematical modeling of financial markets. Numerical methods for solving the Black-Scholes equations include finite difference methods, finite element methods, Monte Carlo methods and Fourier spectral methods [1, 2, 3, 5, 17, 21]. While these methods offer practical advantages and address specific computational requirements, significant challenges remain in tackling high-dimensional multi-asset derivative pricing problems due to the curse of dimensionality.

In this context, quantum computing has emerged as a transformative solution for financial mathematics, potentially enabling polynomial or exponential speedups in complex derivative pricing problems like Black-Scholes while improving computational efficiency and accuracy. Rebentrost et al. [16] presented quantum algorithms for Monte Carlo estimation of the Black-Scholes

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equations, achieving a square-order speedup over classical sampling. In addition, Stamatopoulos et al. [19] extended these capabilities using gate-based methods to price diverse options, highlighting quantum computing's versatility for complex financial instruments. Calderer et al. [4] further advanced the field with a quantum circuit based on the one-dollar Black-Scholes model, proving both feasibility and practical advantages for option pricing. These developments underscore quantum computing's significant potential to revolutionize option pricing methodologies.

This paper is based on a quantum simulation algorithm for Black-Scholes equations using the Schrödingerisation method. The technique transforms financial PDEs with non-unitary dynamics into Schrödinger-type systems through a warped phase transformation, enabling efficient quantum solutions to derivative pricing problems. The method's versatility is demonstrated through implementations in both qubit-based [13, 12] and continuous-variable [9] quantum computing frameworks.

Despite significant progress in quantum computing for PDEs, the explicit design of quantum circuits for solving partial differential equations remains a key challenge. Recent breakthroughs include the work of Sato et al. [18], who developed a scalable Bell-basis approach for quantum circuit construction tailored to wave and heat equations. Building on Schrödingerisation techniques, Hu et al. [8] further demonstrated the method's versatility by designing quantum circuits for both heat and advection equations.

In this work, we present quantum circuits for the Black-Scholes equation based on the Schrödingerisation method. Our contributions include:

- Rigorous derivation of error bounds and computational complexity estimates;
- Demonstration of polynomial speedup over classical methods in high-dimensional settings;
- Numerical verification on Qiskit, confirming both convergence properties and theoretical complexity analysis.

The paper is organized as follows: Section 2 introduces fundamental notations. Sections 3 and 4 present quantum circuit implementations for one-dimensional and d-dimensional Black-Scholes equations, respectively. Section 5 provides a complexity analysis, while Section 6 discusses numerical results. Conclusions are drawn in Section 7.

Notation: We use O for upper bounds and $\tilde{O}$ to ignore logarithmic terms. Pauli matrices are denoted by X, Y, and Z.

2 Notations for finite difference operators

Consider the one-dimensional domain $\Omega := (0, L)$ where L is length of the domain. The domain Ω is uniformly subdivided into $N_x = 2^{n_x}$ intervals with the interval $h = L/N_x$. The discrete function $\mathbf{u} := [u_0, u_1, \cdots, u_{Nx-1}]$ can express as a quantum state $|u\rangle := \sum_{j=0}^{2^{n_x}-1} u_j |j\rangle$.

For the qubit-based system, the finite difference operators can be represented as matrix product operators (MPOs) using the following three 2×2 matrices

$$\sigma_{01} := |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \sigma_{10} := |1\rangle\langle 0| = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad I := |0\rangle\langle 0| + |1\rangle\langle 1|.$$

The shift operators can be defined as

$$
\begin{aligned}
S^- &:= \sum_{j=1}^{2^{n_x}-1} |j-1\rangle\langle j| = \sum_{j=1}^{n_x} I^{\otimes(n_x-j)} \otimes \sigma_{01} \otimes \sigma_{10}^{\otimes(j-1)} \triangleq \sum_{j=1}^{n_x} s_j^-, \\
S^+ &:= (S^-)^\dagger = \sum_{j=1}^{2^{n_x}-1} |j\rangle\langle j-1| = \sum_{j=1}^{n_x} I^{\otimes(n_x-j)} \otimes \sigma_{10} \otimes \sigma_{01}^{\otimes(j-1)} \triangleq \sum_{j=1}^{n_x} s_j^+.
\end{aligned} \tag{2.1}
$$

The forward and backward finite difference operators are,

$$
(D^+\mathbf{u})_j = \frac{u_{j+1}-u_j}{h}, \quad (D^+\mathbf{u})_j = \frac{u_j-u_{j-1}}{h}, \quad j = 0, 1, \cdots, N_x - 1. \tag{2.2}
$$

The difference operator with Dirichlet boundary conditions can be expressed as follows,

$$
D_D^+ = \frac{S^- - I^{\otimes n_x}}{h}, \quad D_D^- = \frac{I^{\otimes n_x} - S^+}{h}, \quad D_D^\pm = \frac{S^- - S^+}{2h}, \quad D_D^\triangle = \frac{S^- + S^+ - 2I^{\otimes n_x}}{h^2}.
$$

3 The one-dimensional Black-Scholes equation

In this section, we implement quantum circuits for the one-dimensional Black-Scholes model of European call options using the Schrödingerisation quantum simulation framework.

Consider a scalar field w governed by the one-dimensional Black-Scholes equation,

$$
\frac{\partial w}{\partial t} + rs\frac{\partial w}{\partial s} + \frac{\sigma^2 s^2}{2}\frac{\partial^2 w}{\partial s^2} = rw. \tag{3.1}
$$

This equation is a variable coefficient parabolic convection-diffusion equation. Here, $s \in [s_{\min}, s_{\max}]$ represents the stock price and w is the option price. r and σ are the risk-free interest rate and volatility, respectively, which are typically positive constants.

Through variable substitution $s = e^x, -\infty < x < \infty$ and $\tau = T - t$, equation (3.1) is converted into a positive constant-coefficient partial differential equation

$$
\begin{cases}
\dfrac{\partial w}{\partial \tau} = \left(r - \dfrac{\sigma^2}{2}\right)\dfrac{\partial w}{\partial x} + \dfrac{\sigma^2}{2}\dfrac{\partial^2 w}{\partial x^2} - rw, \\
w(x,0) = w_0,
\end{cases} \tag{3.2}
$$

where $w_0 = w(0,x) = \max(e^x - K, 0)$. The European option contract is mainly considered, and the exercise price is assumed to be K. The Dirichlet boundary conditions are represented as

$$
w(x,\tau) = 0, \quad x \to -\infty, \quad w(x,\tau) = e^x - Ke^{-r\tau}, \quad x \to +\infty.
$$

Alternatively, we may impose the following mixed boundary conditions,

$$
w(x,\tau) = 0 \quad \text{as} \quad x \longrightarrow -\infty, \quad \frac{\partial w(x,\tau)}{\partial x} = e^x \quad \text{as} \quad x \longrightarrow \infty,
$$

which remain consistent with the Dirichlet boundary condition formulation of model (3.2).

To facilitate the application of the Schrödingerisation method, we introduce the variable substitution $u = w - e^x$, which transforms both the equation and the Dirichlet boundary conditions

to:

$$\begin{cases} \dfrac{\partial u}{\partial \tau} = \left(r - \dfrac{\sigma^2}{2}\right)\dfrac{\partial u}{\partial x} + \dfrac{\sigma^2}{2}\dfrac{\partial^2 u}{\partial x^2} - ru, \\ u(x,0) = w_0 - e^x, \\ u(x,\tau) = -e^x, \quad x \to -\infty, \\ u(x,\tau) = -Ke^{-r\tau}, \quad x \to +\infty. \end{cases} \tag{3.3}$$

Assuming that $x_{\min}$ is a sufficiently small negative number and $x_{\max}$ is a sufficiently large positive number, let the interval $[L, R]$ be the truncation of the logarithmic space of asset prices which can be uniformly subdivided into $N_x = 2^{n_x}+1$ intervals with the interval $h = (R-L)/N_x$. The equation (3.3) can be solved using the central difference method in classical simulation as follows:

$$\frac{d\mathbf{u}}{d\tau} = \left(r - \frac{\sigma^2}{2}\right) D_D^{\pm}\mathbf{u} + \frac{\sigma^2}{2} D_D^{\triangle}\mathbf{u} - rI\mathbf{u} + \mathbf{b} \triangleq A\mathbf{u} + \mathbf{b}, \quad \mathbf{u}(0) = \mathbf{u}_0, \tag{3.4}$$

where $\mathbf{u} = [u(x_1,\tau), u(x_2,\tau), \cdots, u(x_{N_x-1},\tau)]^T$, and $u(x_0,\tau) = -e^{x_0}$, $u(x_{N_x},\tau) = -Ke^{-r\tau}$. The initial value is given by

$$\mathbf{u}_0 = [u(x_1,0), u(x_2,0), \cdots, u(x_{N_x-1},0)]^T$$

and

$$\mathbf{b} = \left[\left(-\frac{r-\frac{\sigma^2}{2}}{2h} + \frac{\sigma^2}{2h^2}\right) u(x_0,\tau), 0, \cdots, 0, \left(\frac{r-\frac{\sigma^2}{2}}{2h} + \frac{\sigma^2}{2h^2}\right) u(x_{N_x},\tau)\right]^T. \tag{3.5}$$

Remark 3.1. *For mixed boundary conditions, the internal nodes are treated in the same way as the Dirichlet condition, only virtual point $x_{N_x+1} = x_{N_x} + h$ need to be introduced at the right boundary of the interval. The first derivative is discretized by central difference method*

$$\frac{u_{N_x+1} - u_{N_x-1}}{2h} = 0.$$

It is reasonable to assume the above grid discretization at x_{N_x}, then

$$\frac{du_{N_x}}{dt} = \left(-\frac{\sigma^2}{h^2} - r\right) u_{N_x} + \left(\frac{r-\frac{\sigma^2}{2}}{2h} + \frac{\sigma^2}{2h^2}\right) u_{N_x+1} + \left(-\frac{r-\frac{\sigma^2}{2}}{2h} + \frac{\sigma^2}{2h^2}\right) u_{N_x-1}.$$

Then with (3.4), the ODE system with mixed boundary conditions is obtained. Its matrix dimension is one order higher than the discrete matrix A of the Dirichlet boundary condition model.

Remark 3.2. *In Equation (3.5), since x_0 is a sufficiently small negative number, we can take the boundary condition as $u(x_0,\tau) = 0$, i.e., $\mathbf{b} = \left[0, 0, \cdots, 0, \left(\frac{r-\frac{\sigma^2}{2}}{2h} + \frac{\sigma^2}{2h^2}\right) u(x_{N_x},\tau)\right]^T$.*

3.1 Schrödingerisation

Introduce the auxiliary variable $\mathbf{R}(t)$ and let $\mathbf{R}(0) = (1, \cdots, 1)^T$, $B = \mathrm{diag}(0, \cdots, 0, -K(\frac{1}{2}\frac{\sigma^2}{h^2} + \frac{r-\frac{\sigma^2}{2}}{2h}))$. By dilating the system $\mathbf{u} \to \bar{\mathbf{u}} = \mathbf{u} \otimes |0\rangle + \mathbf{R}(t) \otimes |1\rangle$, one can derive

$$\frac{d\bar{\mathbf{u}}}{d\tau} = C\bar{\mathbf{u}}, \quad C = \begin{bmatrix} A & B \\ \mathbf{0} & -rI \end{bmatrix}. \tag{3.6}$$

In this case, $C \neq C^\dagger$, we decompose the matrix as

$$\begin{aligned}
&A = A_1 + iA_2, \quad C = C_1 + iC_2,\\
&C_1 = (C + C^\dagger)/2 = C_1^\dagger, \quad C_2 = (C - C^\dagger)/2i = C_2^\dagger,\\
&C_1 = \begin{bmatrix} A_1 & \frac{1}{2}B \\ \frac{1}{2}B & rI \end{bmatrix} = |0\rangle\langle 0| \otimes A_1 + |1\rangle\langle 1| \otimes rI + X \otimes \frac{1}{2}B,\\
&C_2 = \begin{bmatrix} A_2 & \frac{1}{2i}B \\ -\frac{1}{2i}B & \mathbf{0} \end{bmatrix} = |0\rangle\langle 0| \otimes A_2 + Y \otimes \frac{1}{2}B.
\end{aligned} \tag{3.7}$$

Applying the Schrödingerisation method with the warped phase transformation $\mathbf{v}(\tau, p) = e^{-p}\bar{\mathbf{u}}(\tau)$, $p > 0$, we obtain

$$\partial_\tau \mathbf{v}(\tau, p) = C_1 \mathbf{v}(\tau, p) + iC_2 \mathbf{v}(\tau, p) = -C_1 \partial_p \mathbf{v}(\tau, p) + iC_2 \mathbf{v}(\tau, p), \quad p > 0, \tag{3.8}$$

and this equation is extended to $p < 0$ with initial data $\mathbf{v}(0, p) = e^{-|p|}\bar{\mathbf{u}}(0)$. Fourier transform with respect to p leads to

$$\partial_\tau \hat{\mathbf{v}}(\tau, \eta) = i(\eta C_1 + C_2)\hat{\mathbf{v}}(\tau, \eta) \triangleq iH_{BS}\hat{\mathbf{v}}(\tau, \eta),$$

with H_{BS} being Hermitian.

Remark 3.3. *For cash or worthless options with Dirichlet or mixed boundary conditions where the option price is equal to 0 at the boundaries, satisfying* $\mathbf{b} = 0$*, the Schrödingerisation method can be directly applied to equation* (3.4) *without dilation. By directly converting* A *into a Hermitian matrix, the Schrödingerisation method can be used.*

3.2 Discretization

Discretize the domain in p as $-\pi L_p = p_0 < \cdots < p_{N_p} = \pi L_p$ with a mesh size of $\Delta p = 2\pi L_p / N_p$. Define $\eta_k = \left(k - \frac{N_p}{2}\right)/L_p$ for $k = 0, 1, \cdots, N_p - 1$. The variables $\mathbf{v}$ and $\hat{\mathbf{v}}$ are discretized as $\mathbf{v}(t) := [v(x_j, p_k, t)]_{j,k}$ and $\hat{\mathbf{v}}(t) := [\hat{v}(x_j, \eta_k, t)]_{j,k}$ with $1 \le j \le N_x - 1$, $0 \le k \le N_p - 1$. The initial conditions can be represented as

$$\begin{aligned}
&\mathbf{v}(0) := \bar{\mathbf{u}}(0) \otimes \mathbf{p} = [\bar{u}(x_1, 0), \bar{u}(x_2, 0), \cdots, \bar{u}(x_{2N_x-2}, 0)] \otimes [e^{-|p_0|}, e^{-|p_1|}, \cdots, e^{-|p_{N_p-1}|}],\\
&\hat{\mathbf{v}}(0) := \bar{\mathbf{u}}(0) \otimes \eta = [\bar{u}(x_1, 0), \bar{u}(x_2, 0), \cdots, \bar{u}(x_{2N_x-2}, 0)] \otimes \left[\frac{1}{\eta_0^2 + 1}, \frac{1}{\eta_1^2 + 1}, \cdots, \frac{1}{\eta_{N_p-1}^2 + 1}\right].
\end{aligned} \tag{3.9}$$

The Hamiltonian H_{BS} is discretized as

$$
\begin{aligned}
H_{BS} &= C_1 \otimes D_\eta + C_2 \otimes I^{\otimes n_p} \\
&= \left(|0\rangle\langle 0| \otimes \left(\frac{\sigma^2}{2} D_D^{\Delta} - r I^{\otimes n_x} \right) - |1\rangle\langle 1| \otimes r I^{\otimes n_x} + X \otimes \frac{1}{2} B \right) \otimes D_\eta \\
&+ \left(|0\rangle\langle 0| \otimes \frac{1}{i} \left(r - \frac{\sigma^2}{2} \right) D_D^{\pm} + Y \otimes \frac{1}{2} B \right) \otimes I^{\otimes n_p} \\
&= \sum_{k=0}^{N_p-1} \left(k - \frac{N_p}{2} \right) \left(|0\rangle\langle 0| \otimes (\frac{\sigma^2}{2} H_1 - \frac{r}{L_p} I^{\otimes n_x}) - |1\rangle\langle 1| \otimes \frac{r}{L_p} I^{\otimes n_x} \right) \otimes |k\rangle\langle k| \\
&+ \sum_{k=0}^{N_p-1} \left(k - \frac{N_p}{2} \right) / L_p \left(X \otimes \frac{1}{2} B \right) \otimes |k\rangle\langle k| \\
&+ \left(|0\rangle\langle 0| \otimes \left(r - \frac{\sigma^2}{2} \right) H_2 + Y \otimes \frac{1}{2} B \right) \otimes I^{\otimes n_p},
\end{aligned}
\tag{3.10}
$$

where

$$
\begin{aligned}
H_1 &= \gamma_1 \left[\sum_{j=1}^{n_x} (s_j^- + s_j^+) - 2 I^{\otimes n_x} \right], \quad \gamma_1 = \frac{1}{h^2 L_p}, \\
H_2 &= -i\gamma_2 \left[\sum_{j=1}^{n_x} (s_j^- - s_j^+) \right], \quad \gamma_2 = \frac{1}{2h}.
\end{aligned}
\tag{3.11}
$$

3.3 Quantum circuit

By employing the first-order Lie-Trotter-Suzuki decomposition, we can approximate the time evolution operator $U_{BS}(\tau) := \exp(i\tau H_{BS})$,

$$
\begin{aligned}
&U_{BS}(\tau) \\
&\approx \exp\left(i\tau \left(|0\rangle\langle 0| \otimes \left(r - \frac{\sigma^2}{2} \right) H_2 + Y \otimes \frac{1}{2} B \right) \otimes I^{\otimes n_p} \right) \\
&\exp\left(i\tau \sum_{k=0}^{N_p-1} \left(k - \frac{N_p}{2} \right) \left(|0\rangle\langle 0| \otimes (\frac{\sigma^2}{2} H_1 - \frac{r}{L_p} I^{\otimes n_x}) - |1\rangle\langle 1| \otimes \frac{r}{L_p} I^{\otimes n_x} \right) \otimes |k\rangle\langle k| \right) \\
&\exp\left(i\tau \sum_{k=0}^{N_p-1} \left(k - \frac{N_p}{2} \right) / L_p \left(X \otimes \frac{1}{2} B \right) \otimes |k\rangle\langle k| \right) \\
&= \left(|0\rangle\langle 0| \otimes U_2 \left(\left(r - \frac{\sigma^2}{2} \right) \tau \right) + |1\rangle\langle 1| \otimes I^{\otimes n_x} \right) U_2^{(1)}(\tau) \otimes I^{\otimes n_p} \\
&\cdot \sum_{k=0}^{N_p-1} \left(\left(|0\rangle\langle 0| \otimes \mathrm{Ph}(-\tau \frac{r}{L_p}) U_1 \left(\frac{\sigma^2}{2} \tau \right) + |1\rangle\langle 1| \otimes \mathrm{Ph}(-\tau \frac{r}{L_p}) I^{\otimes n_x} \right) U_1^{(1)}(\tau) \right)^{k - N_p/2} \otimes |k\rangle\langle k| \\
&\triangleq U_*(\tau),
\end{aligned}
\tag{3.12}
$$

where

$$
\begin{aligned}
U_j(\tau) &:= \exp\left(i\tau H_j\right), \quad j=1,2,\\
U_1^{(1)}(\tau) &:= \exp\left(i\tau/L_p\left(X\otimes\frac{1}{2}B\right)\right),\\
U_2^{(1)}(\tau) &:= \exp\left(i\tau Y\otimes\frac{1}{2}B\right),\\
\mathrm{Ph}(\theta) &:= e^{i\theta}I^{\otimes n_x},
\end{aligned}
\tag{3.13}
$$

and we have used the following properties:

$$
\exp\left(\sum_k A_k\otimes|k\rangle\langle k|\right)=\sum_k \exp(A_k)\otimes|k\rangle\langle k|.
$$

Noticing that B is an $n\times n$ diagonal matrix with only one nonzero element at the (n,n) position, and $U_1^{(1)}$ and $U_2^{(1)}$ can be simulated by multi-controlled $R_x,\ R_y$ gates. Again using the first-order Lie-Trotter-Suzuki decompositon, $U_1(\tau)$ is approximated as

$$
\begin{aligned}
U_1(\tau) &= \exp\left(i\tau\gamma_1\left[\sum_{j=1}^{n_x}(s_j^- + s_j^+) - 2I^{\otimes n_x}\right]\right)\\
&= \exp(-2i\gamma_1\tau)\exp\left(i\tau\gamma_1\left[\sum_{j=1}^{n_x}\left(s_j^- + s_j^+\right)\right]\right)\\
&\triangleq \mathrm{Ph}(-2\gamma_1\tau)U_1^{(2)}(\tau).
\end{aligned}
\tag{3.14}
$$

U_2 is approximated as

$$
U_2(\tau)=\exp\left(i\tau\gamma_2\left[\sum_{j=1}^{n_x}-i(s_j^- - s_j^+)\right]\right)\triangleq U_2^{(2)}(\tau).
\tag{3.15}
$$

According to [8], one can utilize an explicit quantum circuit which solves the general form $\gamma\sum_{j=1}^{n_x}(e^{i\lambda}s_j^- + e^{-i\lambda}s_j^+)$ with $\lambda,\gamma\in\mathbb{R}$,

$$
e^{i\lambda}s_j^- + e^{-i\lambda}s_j^+ = I^{\otimes(n_x-j)}\otimes\left(e^{i\lambda}|0\rangle|1\rangle^{\otimes j-1}\langle 1|\langle 0|^{\otimes(j-1)} + e^{-i\lambda}|1\rangle|0\rangle^{\otimes j-1}\langle 0|\langle 1|^{\otimes(j-1)}\right).
$$

Define the unitary matrix Bell basis $B_j(\lambda)$ such that

$$
\begin{aligned}
B_j(\lambda)|0\rangle|1\rangle^{\otimes(n-1)} &= \frac{|0\rangle|1\rangle^{\otimes(j-1)} + e^{-i\lambda}|1\rangle|0\rangle^{\otimes(j-1)}}{\sqrt{2}},\\
B_j(\lambda)|1\rangle|1\rangle^{\otimes(n-1)} &= \frac{|0\rangle|1\rangle^{\otimes(j-1)} + e^{-i\lambda}|1\rangle|0\rangle^{\otimes(j-1)}}{\sqrt{2}}.
\end{aligned}
$$

Furthermore, $B_j(\lambda)$ is constructed by

$$
B_j(\lambda) := \left(\prod_{m=1}^{j-1}\mathrm{CNOT}_m^j\right)P_j(-\lambda)H_j,
$$

where H_j is the Hadamard gate acting on the j-th qubit. $P_j(\lambda)$ is the Phase gate acting on the j-th qubit as

$$P_j(\lambda) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\lambda} \end{bmatrix}.$$

The time evolution operator is formulated as

$$\begin{aligned} \exp(i\gamma\tau(e^{i\lambda}s_j^- + e^{-i\lambda}s_j^+)) &= I^{\otimes(n_x-j)} \otimes B_j(\lambda)\mathrm{CRZ}_j^{1,\cdots,j-1}(-2\gamma\tau)B_j(\lambda)^\dagger \\ &\triangleq I^{\otimes(n_x-j)} \otimes W_j(\gamma\tau,\lambda), \end{aligned}$$

where $\mathrm{CRZ}_j^{1,\cdots,j-1}(\theta) := \exp(-i\theta Z_j/2) \otimes |1\rangle\langle 1|^{\otimes(j-1)} + I \otimes (I^{\otimes(j-1)} - |1\rangle\langle 1|^{\otimes(j-1)})$ is the multicontroled RZ gate acting on the j-th qubit controlled by $1, \cdots, j-1$-th qubits.

Applying the first-order Lie-Trotter-Suzuki decomposition, we have

$$\exp\left(i\gamma\tau\sum_{j=1}^{n_x}(e^{i\lambda}s_j^- + e^{-i\lambda}s_j^+)\right) \approx \prod_{j=1}^{n_x}\exp(i\gamma\tau(e^{i\lambda}s_j^- + e^{-i\lambda}s_j^+)) = \prod_{j=1}^{n_x} I^{\otimes(n_x-j)} \otimes W_j(\gamma\tau,\lambda).$$

From this point, we set $\lambda = 0$, $\gamma = \gamma_1$ and $\lambda = -\frac{\pi}{2}$, $\gamma = \gamma_2$, respectively, to simulate $U_1^{(2)}$ and $U_2^{(2)}$ in (3.14) and (3.15). Then U_1 and U_2 can be constructed explicitly as

$$\begin{aligned} U_1(\tau) &\approx \mathrm{Ph}(-2\gamma_1\tau)\prod_{j=1}^{n_x} I^{\otimes(n_x-j)} \otimes W_j(\gamma_1\tau, 0) \triangleq V_1(\tau), \\ U_2(\tau) &\approx \prod_{j=1}^{n_x} I^{\otimes(n_x-j)} \otimes W_j(\gamma_2\tau, -\frac{\pi}{2}) \triangleq V_2(\tau). \end{aligned} \tag{3.16}$$

Ultimately, the time evolution operator $U_{BS}(\tau)$ is approximated by

$$V_{BS}(\tau) := \tilde{V}_2(\tau) \otimes I^{\otimes n_p} \sum_{k=0}^{N_p-1} \tilde{V}_1(\tau)^{k-N_p/2} \otimes |k\rangle\langle k|, \tag{3.17}$$

where

$$\begin{aligned} \tilde{V}_1(\tau) &= \left(|0\rangle\langle 0| \otimes \mathrm{Ph}(-\tau\frac{r}{L_p})V_1\left(\frac{\sigma^2}{2}\tau\right) + |1\rangle\langle 1| \otimes \mathrm{Ph}(-\tau\frac{r}{L_p})I^{\otimes n_x}\right)U_1^{(1)}(\tau), \\ \tilde{V}_2(\tau) &= \left(|0\rangle\langle 0| \otimes V_2\left(\left(r - \frac{\sigma^2}{2}\right)\tau\right) + |1\rangle\langle 1| \otimes I^{\otimes n_x}\right)U_2^{(1)}(\tau). \end{aligned}$$

In order to minimize the gate count during the implementation, we utilize the binary representation of integers $k = (k_{n_p-1\cdots k_0.}) = \sum_{m=0}^{n_p-1} k_m 2^m$ to obtain

$$\tilde{V}_{BS}(\tau) = \tilde{V}_2(\tau)\tilde{V}_1(\tau)^{-N_p/2} \otimes I^{\otimes n_p} \prod_{m=0}^{\prime n_p-1} \left(\tilde{V}_1(\tau)^{2^m} \otimes |1\rangle\langle 1| + I^{\otimes n_x} \otimes |0\rangle\langle 0|\right),$$

where the primed product $\prod'$ denotes the regular matrices product for the first register (consisting of n_x+1 qubits) and the tensor product for the second register (consisting of n_p qubits). Assuming that $\tilde{V}_{BS}(\tau)$can be implemented with a cost independent of k, this approach is exponentially more efficient than directly implementing $V_{BS}(\tau)$. The detailed quantum circuits for the approximating unitary matrices are depicted in Figure 3.1-3.4.

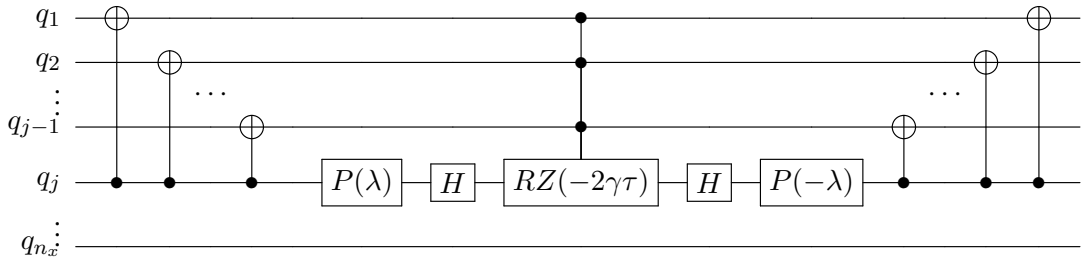


Fig. 3.1. Quantum circuit for $W_j(\gamma\tau, \lambda)$

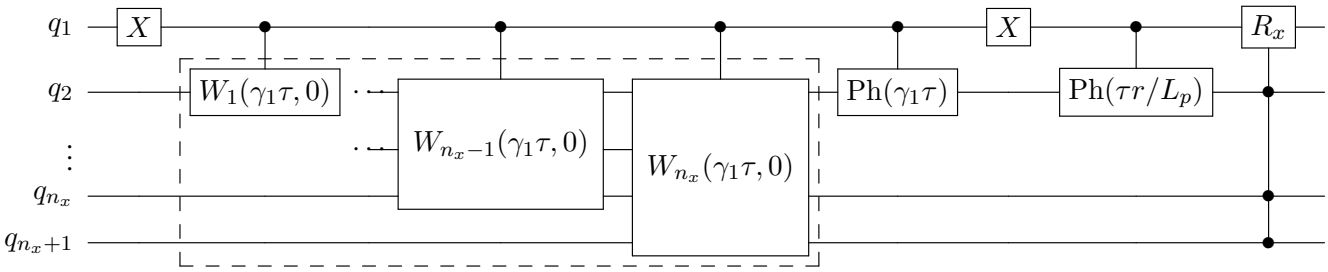


Fig. 3.2. Quantum circuit for $\tilde{V}_1(\tau)$

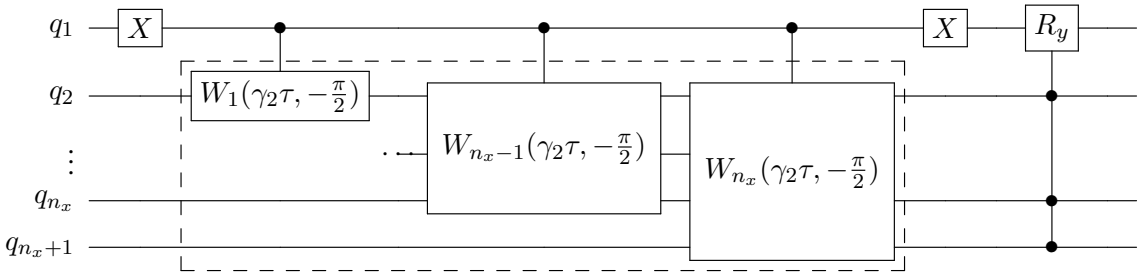


Fig. 3.3. Quantum circuit for $\tilde{V}_2(\tau)$

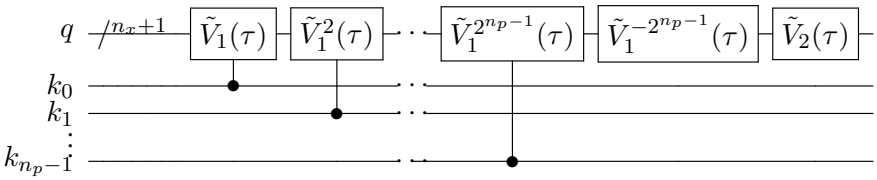


Fig. 3.4. Quantum circuit for $\tilde{V}_{BS}$

For time $T = N\tau$, one can obtain $|\hat{\mathbf{v}}(T)\rangle := V_{BS}^N|\hat{\mathbf{v}}(0)\rangle$, which means the operator $V_{BS}(\tau)$ is applied N times. We can perform an inverse quantum Fourier transform on it to retrieve $|\mathbf{v}(T)\rangle$. Then a measurement (the projection $P := I^{\otimes n_x+1} \otimes |p_k^\diamond\rangle\langle p_k^\diamond|$) is taken to select only the $|p_k^\diamond\rangle$ part of the state $|\mathbf{u}(T)\rangle$. The full circuit to implement the Schrödingerisation method is shown in Figure 3.5.

Remark 3.4. *According to Theorem 5.1 in [10], $p_k^\diamond > \max(\lambda_n(C_1)T, 0)$, where $\lambda_n(C_1)$ is the max eigenvalues of matrix C_1, we can recovery of the original variables $\mathbf{u}$. While $-rI$ is a negative number multiplied by the identity matrix and B in (3.6) is a matrix with only one non-zero entry, we just focus on the eigenvalues of A_1.*

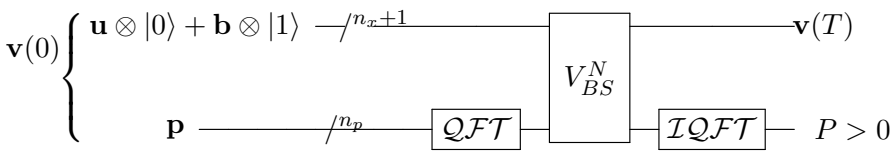


Fig. 3.5. Quantum circuit for the Schrödingerisation method, where the measurement requires only projection onto $P > 0$ and $\mathcal{QFT}(\mathcal{IQFT})$ denotes the (inverse) quantum Fourier transform.

4 The d-dimension Black-Scholes equation

Denote $s_1, s_2, \cdots, s_d$ as the prices of d risky assets. W is the option price based on the underlying assets $s_1, s_2, \cdots, s_d$, and its pricing law satisfies the d-dimensional Black-Scholes equation:

$$\frac{\partial U}{\partial t} + \frac{1}{2}\sum_{m,n=1}^{d} \sigma_m \sigma_n \rho_{mn} s_m s_n \frac{\partial^2 U}{\partial s_m \partial s_n} + \sum_{m=1}^{d} r s_m \frac{\partial U}{\partial s_m} - rU = 0,$$

where $t \in [0, T]$, and $\mathbf{s} = [s_1, \cdots, s_d]^T \in (0, s^1_{\max}) \times \cdots \times (0, s^d_{\max})$, $\rho_{m,n} \in [-1, 1]$, $(m, n = 1, \cdots, d)$.

Using the similar technique applied for the 1d case, set $s_i = e^{x_i}, \tau = T - t$, one can derive:

$$\frac{\partial U}{\partial t} = \frac{1}{2}\sum_{m,n=1}^{d} \sigma_m \sigma_n \rho_{mn} \frac{\partial^2 U}{\partial x_m \partial x_n} + \sum_{m=1}^{d} \left(r - \frac{\sigma_m^2}{2}\right) \frac{\partial U}{\partial x_m} - rU. \tag{4.1}$$

For high dimensional Black-Scholes equation, mixed boundary conditions are often used in application. Assume the following boundary condition for call option

$$U(x_1, \cdots, x_{j-1}, L, x_{j+1}, \cdots, x_d) = 0, \quad \frac{\partial U}{\partial x_j}(x_1, \cdots, x_{j-1}, R, x_{j+1}, \cdots, x_d) = 0.$$

Then the equation (4.1) can be solved using the central difference method as follows:

$$\frac{dU}{d\tau} = \sum_{m=1}^{d} \left(r - \frac{\sigma_m^2}{2}\right) (D_D^{\pm})_m U + \sum_{m,n=1}^{d} \frac{\sigma_m \sigma_n \rho_{mn}}{2} (D_D^{\Delta})_{mn} U - rU \triangleq \mathbf{A}U, \tag{4.2}$$

where $U = [U_{j_1,\cdots,j_d}]_{1 \le j_i \le N_x - 1}$, $(D_D^{\pm})_m = I^{\otimes(m-1)n_x} \otimes D_D^{\pm} \otimes I^{\otimes(d-m-1)n_x}$.

Remark 4.1. *In (4.2), the operator D_D^{Δ} is expressed by the central difference formula*

$$(D_D^{\Delta})_{mn} u = \begin{cases} \frac{1}{h^2}(u_{m+1,n} - 2u_{m,n} + u_{m-1,n}), & m = n, \\ \frac{1}{4h^2}(u_{m+1,n+1} - u_{m+1,n-1} - u_{m-1,n+1} - u_{m-1,n-1}), & m \neq n. \end{cases}$$

For the case $m = n$, one can obtain $(D_D^{\Delta})_{mn} = I^{\otimes(m-1)n_x} \otimes D_D^{\Delta} \otimes I^{\otimes(d-m-1)n_x}$, where D_D^{Δ} is defined in (2.2). while for $m \neq n$, $(D_D^{\Delta})_{mn} = I^{\otimes(m-1)n_x} \otimes \frac{1}{2h} S^+ \otimes I^{\otimes(d-m-n)n_x} \otimes D_D^{\pm} \otimes I^{\otimes(d-n-1)n_x} + I^{\otimes(m-1)n_x} \otimes \frac{1}{2h} S^- \otimes I^{\otimes(d-m-n)n_x} \otimes D_D^{\pm} \otimes I^{\otimes(d-n-1)n_x}$, where S^+ and S^- are shift operators defined in (2.1).

4.1 Schrödingerisation

Most banks use a simplified model for ordinary basket options that ignores weak correlations. Thus, in this section, we consider the simple case $\rho_{mn} = 0, m \neq n$, then the homogeneous ordinary differential equation system (4.2) is

$$\frac{dU}{d\tau} = \mathbf{A}U, \quad U(0) = U_0. \tag{4.3}$$

In this case, $\mathbf{A} \neq -\mathbf{A}^\dagger$, we decompose the matrix as

$$\begin{aligned} &\mathbf{A} = \mathbf{A}_1 + i\mathbf{A}_2, \\ &\mathbf{A}_1 = (\mathbf{A} + \mathbf{A}^\dagger)/2 = \mathbf{A}_1^\dagger, \quad \mathbf{A}_2 = (\mathbf{A} - \mathbf{A}^\dagger)/2i = \mathbf{A}_2^\dagger, \end{aligned}$$

where $\mathbf{A}_i = \sum_{j=1}^d I^{\otimes(j-1)n_x} \otimes A_i \otimes I^{\otimes(d-j)n_x}$, $(i = 1, 2)$, A_i is defined in (3.7).

Applying the Schrödingerisation method, the warped phase transformation $\mathcal{V}(\tau, p) = e^{-p}\bar{U}(\tau)$, $p > 0$, satisfies,

$$\partial_\tau \mathcal{V}(\tau, p) = \mathbf{A}_1\mathcal{V}(\tau, p) + i\mathbf{A}_2\mathcal{V}(\tau, p) = -\mathbf{A}_1\partial_p\mathcal{V}(\tau, p) + i\mathbf{A}_2\mathcal{V}(\tau, p), \quad p > 0,$$

and this equation is extended to $p < 0$ with initial data $\mathcal{V}(0, p) = e^{-|p|}U(0)$. After the Fourier transform with respect to p, we obtain,

$$\partial_\tau \hat{\mathcal{V}}(\tau, \eta) = i(\eta\mathbf{A}_1 + \mathbf{A}_2)\hat{\mathcal{V}}(\tau, \eta) \triangleq i\mathbf{H}_{BS}\hat{\mathcal{V}}(\tau, \eta),$$

with $\mathbf{H}_{BS}$ being Hermitian.

4.2 Quantum circuit

In the d-dimensional case, one has

$$\begin{aligned} \mathbf{H}_{BS} &= \mathbf{A}_1 \otimes D_\eta + \mathbf{A}_2 \otimes I^{\otimes n_p} \\ &= \sum_{k=0}^{N_p-1} \left(k - \frac{N_p}{2}\right) \sum_{m=1}^{d} \left(\frac{\sigma_m^2 \rho_{mm}}{2}(H_1)_m\right) \otimes |k\rangle\langle k| \\ &+ \sum_{m=1}^{d} \left(r - \frac{\sigma_m^2}{2}\right)(H_2)_m \otimes I^{\otimes n_p}, \end{aligned} \tag{4.4}$$

where $(H_1)_{mm} := I^{\otimes(m-1)n_x} \otimes H_1 \otimes I^{\otimes(d-m)n_x}$, $(H_2)_m := I^{\otimes(m-1)n_x} \otimes H_2 \otimes I^{\otimes(d-m)n_x}$, $H_i (i = 1, 2)$ defined in (3.11).

By employing the first-order Lie-Trotter-Suzuki decomposition, we can approximate the time evolution operator $\mathbf{U}_{BS}(\tau) := \exp(i\mathbf{H}_{BS}\tau)$,

$$
\begin{aligned}
\mathbf{U}_{BS}(\tau) \approx & \exp\left(i\tau \sum_{m=1}^{d}\left(r-\frac{\sigma_m^2}{2}\right)(H_2)_m \otimes I^{\otimes n_p}\right) \\
& \exp\left(i\tau \sum_{k=0}^{N_p-1}\left(k-\frac{N_p}{2}\right)\sum_{m=1}^{d}\frac{\sigma_m^2\rho_{mm}}{2}(H_1)_m\right) \otimes |k\rangle\langle k| \\
= & \prod_{m=1}^{d} U_2\left(\left(r-\frac{\sigma_m^2}{2}\right)\tau\right)_m \otimes I^{\otimes n_p} \\
& \sum_{k=0}^{N_p-1}\left(\prod_{m=1}^{d} U_1\left(\frac{\sigma_m^2\rho_{mm}}{2}\tau\right)_m\right)^{k-N_p/2} \otimes |k\rangle\langle k| \\
\triangleq & \mathbf{U}_*(\tau).
\end{aligned}
\tag{4.5}
$$

Similar to the 1-dimensional case, using the first-order Lie-Trotter-Suzuki decomposition, the time evolution operator $\mathbf{U}_{BS}(\tau)$ is approximated by

$$
\mathbf{V}_{BS}(\tau) := \tilde{\mathbf{V}}_2(\tau) \otimes I^{\otimes n_p} \sum_{k=0}^{N_p-1} \tilde{\mathbf{V}}_1(\tau)^{k-N_p/2} \otimes |k\rangle\langle k|, \tag{4.6}
$$

where

$$
\tilde{\mathbf{V}}_1(\tau) = \prod_{m=1}^{d} V_1\left(\frac{\sigma_m^2\rho_{mm}}{2}\tau\right)_m,
$$

$$
\tilde{\mathbf{V}}_2(\tau) = \prod_{m=1}^{d} V_2\left(\left(r-\frac{\sigma_m^2}{2}\right)\tau\right)_m,
$$

and $V_i(\tau)_m = I^{\otimes(m-1)n_x} \otimes V_i \otimes I^{\otimes(d-m)n_x}$ with $V_i, (i=1,2)$ defined in (3.16).

Furthermore,we utilize the binary representation of integers $k = (k_{n_p-1\cdots k_0.}) = \sum_{m=0}^{n_p-1} k_m 2^m$ to minimize the gate count

$$
\mathbf{V}_{BS}(\tau) = (\tilde{\mathbf{V}}_2(\tau)\tilde{\mathbf{V}}_1(\tau)^{-N_p/2} \otimes I^{\otimes n_p})\prod\nolimits_{m=0}^{n_p-1}\left(\tilde{\mathbf{V}}_1(\tau)^{2^m} \otimes |1\rangle\langle 1| + I^{\otimes dn_x} \otimes |0\rangle\langle 0|\right).
$$

For the d-dimension Black-Scholes equation, the detailed quantum circuits for the approximating unitary matrices are depicted in Figure 4.1-4.3. The full circuit to implement the Schrödingerisation method is shown in Figure 3.5 with $dn_x + n_p$ qubits.

Remark 4.2. *For the case $\rho_{mn} \neq 0, m \neq n$, we notice that the Schrödingerisation method is still applicable. While using the Lie-Trotter-Suzuki decomposition, the decomposition of the coupled quantum gates $\exp(S^+ \otimes D_D^{\pm})$ should be considered as discussed in Remark 4.1.*

5 Complexity analysis

In this subsection, we estimate the complexity of the quantum circuits of the one-dimensional and d-dimensional Black-Scholes equation constructed in the previous sections and demonstrate their scalability.

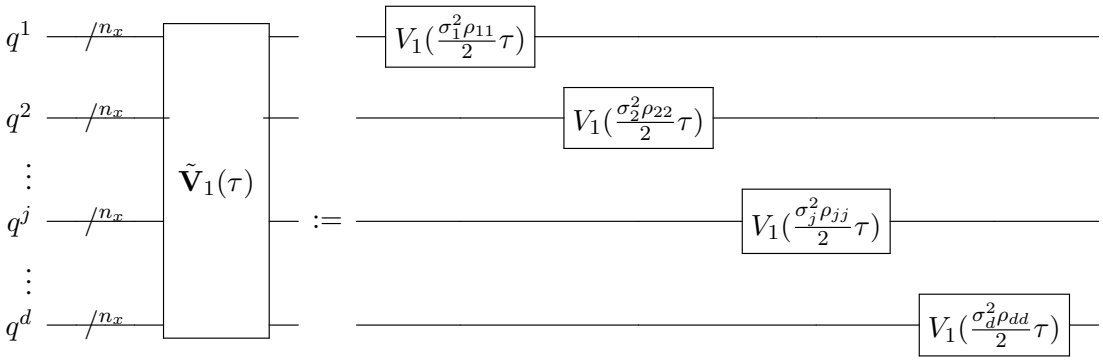


Fig. 4.1. Quantum circuit for $\tilde{\mathbf{V}}_1(\tau)$

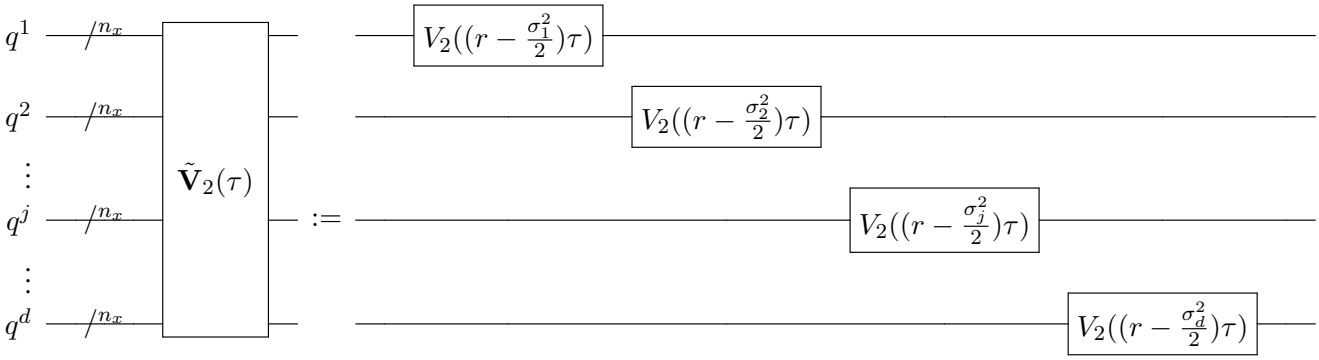


Fig. 4.2. Quantum circuit for $\tilde{\mathbf{V}}_2(\tau)$

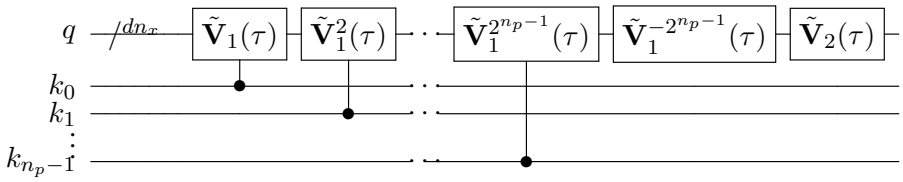


Fig. 4.3. Quantum circuit for $\tilde{\mathbf{V}}_{BS}$

Lemma 5.1. *For the Schrödinger equation* $\frac{d|\mathbf{u}(t)\rangle}{dt} = iH_{BS}|\mathbf{u}(t)\rangle$*, with the Hamiltonian* H_{BS} *as specified in Equation (3.10), the time evolution operator* $U_{BS}(\tau) = \exp(iH_{BS}\tau)$ *over a time step* τ *can be effectively approximated by the unitary operator* $V_{BS}(\tau)$*, which is detailed in Equation (3.17). The quantum circuit realization of* $V_{BS}(\tau)$ *is depicted in Figures 3.1 - 3.4. In addition, the approximation error evaluated in the operator norm, is bounded by*

$$\|U_{BS}(\tau) - V_{BS}(\tau)\| \leq \frac{1}{4}\tau^2 n_x(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)c^2, \tag{5.1}$$

where $N_p = 2^{n_p}$ *and* $N_x - 1 = 2^{n_x}$ *correspond to the number of grid points for the variables* p *and* x*, respectively. The parameters* γ_1 *and* γ_2 *are defined in Equation (3.11), and* c *is given by* $c = \max\left(\frac{\sigma^2}{2}, r - \frac{\sigma^2}{2}\right)$.

Proof. Recalling the definition of $U_1(\tau), U_2(\tau)$, in equation (3.14) and (3.15), $V_1(\tau), V_2(\tau)$ in equation (3.16). According to the theory of the Trotter splitting error with commutator scaling [6] and the analysis in [18], we have

$$
\begin{aligned}
&\|U_1(\tau)-V_1(\tau)\| \\
=&\left\|\exp\left(i\tau\gamma_1\sum_{j=1}^{n_x}\left(s_j^-+s_j^+\right)\right)-\prod_{j=1}^{n_x}I^{\otimes(n_x-j)}\otimes W_j(\gamma_1\tau,0)\right\| \\
\leq&\frac{\gamma_1^2\tau^2}{2}\sum_{j=1}^{n_x}\sum_{j'=j+1}^{n_x}\|[(s_j^-+s_j^+),(s_{j'}^-+s_{j'}^+)]\|\leq\frac{\gamma_1^2\tau^2(n_x-1)}{2},
\end{aligned}
$$

where for the last inequality, we have applied the commutator results (see e.g.[8])

$$
\sum_{j=1}^{n_x}\sum_{j'=j+1}^{n_x}\|[(s_j^-+s_j^+),(s_{j'}^-+s_{j'}^+)]\|=n_x-1.
$$

Similarly, we have

$$
\|U_2(\tau)-V_2(\tau)\|\leq\frac{\gamma_2^2\tau^2(n_x-1)}{2}.
$$

The Trotter error of the first decomposition equation (3.12) is

$$
\begin{aligned}
&\|U_{BS}(\tau)-U_*(\tau)\| \\
\leq&\frac{1}{2}\left\|\left[\sum_{k=0}^{N_p-1}\left(k-\frac{N_p}{2}\right)|0\rangle\langle 0|\otimes H_1\otimes|k\rangle\langle k|,|0\rangle\langle 0|\otimes H_2\otimes I^{\otimes n_p}\right]\right\| \\
\leq&\frac{1}{2}N_p\gamma_1\gamma_2 n_x\left(r-\frac{\sigma^2}{2}\right)\frac{\sigma^2}{2}.
\end{aligned}
$$

The error between U_* and V_{BS} is

$$
\begin{aligned}
&\|U_*(\tau)-V_{BS}(\tau)\| \\
\leq&\left\|U_2\left(\left(r-\frac{\sigma^2}{2}\right)\tau\right)-\tilde{V}_2(\tau)\right\|+\max_{0\leq k\leq N_p-1}\left|k-\frac{N_p}{2}\right|\left\|U_1\left(\frac{\sigma^2}{2}\right)-\tilde{V}_1(\tau)\right\| \\
\leq&\frac{\gamma_2^2\tau^2 n_x}{2}\left(r-\frac{\sigma^2}{2}\right)^2+\frac{N_p}{2}\frac{\gamma_1^2\tau^2 n_x}{2}\left(\frac{\sigma^2}{2}\right)^2.
\end{aligned}
$$

To sum up, we have

$$
\begin{aligned}
\|U_{BS}(\tau)-V_{BS}(\tau)\|\leq&\|U_{BS}(\tau)-U_*(\tau)\|+\|U_*(\tau)-V_{BS}(\tau)\| \\
\leq&\frac{1}{4}\tau^2 n_x(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2,
\end{aligned}
$$

where $c=\max\left(\frac{\sigma^2}{2},r-\frac{\sigma^2}{2}\right)$. □

Lemma 5.2. *The approximated time evolution operator $V_{BS}(\tau)$ in (3.17) can be implemented using $\mathcal{Q}_{single}=O(N_p n_x)$ single-qubit gates and at most $\mathcal{Q}_{V_{BS}}=O(N_p n_x^2)$* CNOT *gates for $n_x\geq 3$.*

Proof. There are a $\tilde{V}_2(\tau)$ gate, a maximum of $2^{n_p-1}\tilde{V}_1$ gates and $\sum_{m=0}^{n_p-1}2^m=2^{n_p}-1$ controlled $\tilde{V}_1$ gates in $V_{BS}(\tau)$. Furthermore, $\tilde{V}_1$ consists of 2 phase gate Ph, W_j, $(j=1,\cdots,n_x)$, and a multi-controlled R_x gate. $\tilde{V}_2$ consists of W_j, $(j=1,\cdots,n_x)$, and a multi-controlled R_y gate. W_j can be decomposed into a multi-controlled RZ gate (a controlled RZ gate for $j=2$), 2 Hadmard

gates and $2(j-1)$ CNOT gates in the case $\lambda = 0$. while $\lambda \neq 0$, decomposition of W_j have two more P gates.

Noticing that we need two more X gates to transfer the control state from $|0\rangle$ to $|1\rangle$ and transfer back. Hence the number of single-qubit gates is

$$\mathcal{Q}_{single} = 2^{n_p-1}(2n_x + 2n_x + 2) + 2n_x + 2n_x + 2 = O(N_p n_x).$$

As for CNOT gates, according to [18] [20], a multicontrolled RZ or RX gate with $(j-1)$ control qubits can be decomposed into single-qubit gates and at most $(16j-40)$ CNOT gates. Thus, we have,

$$\mathcal{Q}_{V_2} = \mathcal{Q}_{V_1} = 2+\sum_{j=3}^{n_x+1}(16j-40)+2n_x+2n_x+\sum_{j=3}^{n_x}2(j-1)*8+16(n_x+1)-40 = 16n_x^2-4n_x-14.$$

The controlled V_1 gate consists of a controlled phase gate and controlled W_j, $(j = 1, \cdots, n_x)$. $c-W_j$ can be decomposed into a multi-controlled RZ gate (a controlled RZ gate for $j = 1$), two controlled H gates, and $2(j-1)$ controlled CNOT gates. Therefore, the CNOT gates in controlled V_1 gate can be estimated by

$$\mathcal{Q}_{c-V_1} = 2+\sum_{j=3}^{n_x+2}(16j-40)+2n_x+2n_x+\sum_{j=3}^{n_x}2(j-1)*24 = 16(n_x+2)-40+2 = 32n_x^2-12n_x-66.$$

Then, the CNOT gates in V_{BS} can be estimated

$$\mathcal{Q}_{BS} = \mathcal{Q}_{V_2} + 2^{n_p-1}\mathcal{Q}_{V_1} + (2^{n_p}-1)\mathcal{Q}_{c-V_1} = O(N_p n_x^2).$$

□

Lemma 5.3. *For time T, the time evolution operator $U_{BS}(T) = \exp(iH_{BS}T)$ can be implemented on a (n_x+1+n_p)-qubit system using quantum circuits with $O(N_pT^2n_x^2(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2/\varepsilon)$ single-qubit gates and $O(N_pn_x^3T^2(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2/\varepsilon)$ CNOT gates, within an additive error of ε.*

Proof. The time interval T is partitioned into N subintervals, with $N = T/\tau$. It is assumed that there exists a small ε,

$$\|U_{BS}^N(\tau) - V_{BS}^N(\tau)\| \leq N\|U_{BS}(\tau) - V_{BS}(\tau)\| \leq \varepsilon. \tag{5.2}$$

With the results (5.1), we have

$$\frac{1}{4N}T^2n_x(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2 \leq \varepsilon,$$

which can be rearranged as

$$N \geq \frac{1}{4\varepsilon}T^2n_x(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2.$$

According to Lemma 5.2, V_{BS} consists of $O(N_pn_x)$ single-qubit gates and at most $O(N_pn_x^2)$ CNOT gates. Therefore, V_{BS}^N can be implemented using $O(N_pT^2n_x^2(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2/\varepsilon)$ single-qubit gates and at most $O(N_pn_x^3T^2(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)c^2/\varepsilon)$ CNOT gates. □

Theorem 5.1. *For the one-dimensional Black-Scholes equation (3.3), the state $|\mathbf{u}(t)\rangle$, where $\mathbf{u}(t)$ is the solution of (3.4) with a mesh size h, can be prepared with precision ε' using the Schrödingerisation method depicted in Figure 3.4 and Figure 3.5. This preparation can be achieved using at most $\tilde{O}\left(\frac{T^2\|\mathbf{u}(0)\|^3}{h^4\varepsilon^3\|\mathbf{u}(T)\|^3}\right)$ single-qubit gates and* CNOT *gates.*

Proof. As illustrated in Figure 3.5, the quantum circuit consists of a quantum Fourier transform (QFT), an inverse quantum Fourier transform (IQFT), a unitary operation $V_{BS}^N(\tau)$, and a projection P.

First, the (inverse) quantum Fourier transform can be implemented using $O(n_p^2)$ controlled rotation (CR) gates, which corresponds to $O(n_p^2)$ CNOT gates [14, 15, 20].

Second, the complexity analysis for simulating the unitary $V_{BS}^N(\tau)$, as concluded in Lemma 5.3, can be applied with $n_x = O(\log(L_x/h))$ and $\gamma_1 = \frac{1}{h^2 L_p}, \gamma_2 = \frac{1}{2h}$. This results in a computational cost of $O\left(\frac{T^2 N_p^2 \log^3(L_x/h)c^2}{h^4 L_p^2 \varepsilon'}\right)$, where ε' denotes the desired precision $|U_{BS}^N(T) - V_{BS}^N(\tau)| \leq \varepsilon'$.

Third, the number of measurements and the error of the output state can be discussed in the context of the given expressions $|\hat{\mathbf{v}}(T)\rangle = U_{BS}^N(T)|\hat{\mathbf{v}}(0)\rangle$ and $|\hat{\mathbf{v}}_D(T)\rangle = V_{BS}^N(T)|\hat{\mathbf{v}}(0)\rangle$. To retrieve $|\hat{\mathbf{v}}(T)\rangle$ and $|\hat{\mathbf{v}}_D(T)\rangle$, an inverse quantum Fourier transform can be performed on these states, followed by a projection P onto $|p_k^\diamond\rangle$ through measurement. Given the relationship $v(t,x,p) = e^{-p}u(t,x)$ for $p > 0$, one can observe that

$$\frac{\|P\mathbf{v}(T) - e^{-p_k^\diamond}\mathbf{u}(T)|p_k^\diamond\rangle\|}{\|e^{-p_k^\diamond}\mathbf{u}(T)\|} = O\left(\frac{\pi L_p}{N_p} + e^{-\pi L_p}\right),$$

which represents the (relative) discretization error over the p variable. It follows by,

$$\begin{aligned}
&\left\|\frac{e^{p_k^\diamond}\|\mathbf{v}(0)\|}{\|\mathbf{u}(T)\|}P\,|\mathbf{v}_D(T)\rangle - |\mathbf{u}(T)\rangle\,|p_k^\diamond\rangle\right\| \\
\leq &\left\|\frac{e^{p_k^\diamond}\|\mathbf{v}(0)\|}{\|\mathbf{u}(T)\|}P\,|\mathbf{v}_D(T)\rangle - \frac{e^{p_k^\diamond}\|\mathbf{v}(0)\|}{\|\mathbf{u}(T)\|}P\,|\mathbf{v}(T)\rangle\right\| + \left\|\frac{e^{p_k^\diamond}\|\mathbf{v}(0)\|}{\|\mathbf{u}(T)\|}P\,|\mathbf{v}(T)\rangle - |\mathbf{u}(T)\rangle\,|p_k^\diamond\rangle\right\| \\
= &O\left(\frac{e^{p_k^\diamond}\|\mathbf{u}(0)\|}{\|\mathbf{u}(T)\|}\varepsilon' + \frac{\pi L_p}{N_p} + e^{-\pi L_p}\right).
\end{aligned}$$

To ensure the precision of the output state, the following conditions are imposed: $L_p = O(\log(1/\varepsilon))$, $N_p = (L_p/\varepsilon) = \tilde{O}(1/\varepsilon)$, $p_k^\diamond = O(1)$, $\varepsilon' = O(\|\mathbf{u}(T)\|\varepsilon/\|\mathbf{u}(0)\|)$. These conditions are essential for maintaining the desired accuracy in the quantum simulation process. The probability of obtaining the desired state is given by $O(\|\mathbf{u}(T)\|^2/\|\mathbf{u}(0)\|^2)$. Consequently, the number of measurements required to achieve the desired state with high probability is $O(\|\mathbf{u}(0)\|^2/\|\mathbf{u}(T)\|^2)$. This ensures that the output state is accurately retrieved with the specified precision.

Combining all the steps, the Schrödingerisation method requires at most $\tilde{O}\left(\frac{T^2\|\mathbf{u}(0)\|^3}{h^4\varepsilon^3\|\mathbf{u}(T)\|^3}\right)$ single-qubit gates and CNOT gates. □

We now extend the above discussion to the operators acting on a d-dimensional domain Ω for special cases $\rho_{mn} = 0, m \neq n$.

Lemma 5.4. *For the d-dimensional Schrödinger equation $\frac{d|U(t)\rangle}{dt} = i\mathbf{H}_{BS}|U(t)\rangle$, the Hamiltonian $\mathbf{H}_{BS}$ is specified by Equation (4.4). The time evolution operator $\mathbf{U}_{BS}(\tau) = \exp(i\mathbf{H}_{BS}\tau)$ with a time increment τ is approximated by the unitary operator $\mathbf{V}_{BS}(\tau)$ as detailed in Equation (4.5). The explicit quantum circuit realization of $\mathbf{V}_{BS}(\tau)$ is illustrated in Figures 4.1–4.3. Additionally, the upper bound on the approximation error, assessed in the operator norm, is given by*

$$\|\mathbf{U}_{BS}(\tau)-\mathbf{V}_{BS}(\tau)\|\leq \frac{1}{4}\tau^2 n_x(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)\sum_{m=1}^{d}c_m^2, \tag{5.3}$$

where d denotes the spatial dimension, $N_p=2^{n_p}$ and $N_x-1=2^{n_x}$ represent the number of grid points for the variables p and x, respectively. Additionally, γ_1,γ_2 are defined in (3.11) and $c_m=\max_m\left(r-\frac{\sigma_m^2}{2},\frac{\sigma_m^2\rho_{mm}}{2}\right)$.

Proof. Similar to the approximation error in one-dimension concluded in Lemma 5.1, for d-dimension case, we have

$$\begin{aligned}\left\|\prod_{m=1}^{d}\left(U_2\left(\left(r-\frac{\sigma_m^2}{2}\right)\tau\right)_m-\tilde{\mathbf{V}}_2(\tau)_m\right)\right\|\leq&\sum_{m=1}^{d}\left\|\left(U_2\left(\left(r-\frac{\sigma_m^2}{2}\right)\tau\right)_m-\tilde{\mathbf{V}}_2(\tau)_m\right)\right\|\\ \leq&\frac{\gamma_2^2\tau^2(n_x-1)}{2}\sum_{m=1}^{d}\left(r-\frac{\sigma_m^2}{2}\right),\end{aligned}$$

$$\begin{aligned}\left\|\prod_{m,n=1}^{d}\left(U_1\left(\frac{\sigma_m^2\rho_{mm}}{4}\tau\right)_m-\tilde{\mathbf{V}}_1(\tau)_m\right)\right\|\leq&\sum_{m=1}^{d}\left\|\left(U_1\left(\frac{\sigma_m^2\rho_{mm}}{4}\tau\right)_m-\tilde{\mathbf{V}}_1(\tau)_m\right)\right\|\\ \leq&\frac{\gamma_1^2\tau^2(n_x-1)}{2}\sum_{m=1}^{d}\frac{\sigma_m^2\rho_{mm}}{4}.\end{aligned}$$

The Trotter error of the first decomposition Equation (4.5) is

$$\begin{aligned}&\|\mathbf{U}_{BS}(\tau)-\mathbf{U}_*(\tau)\|\\ \leq&\frac{1}{2}\left\|\left[\sum_{k=0}^{N_p-1}\left(k-\frac{N_p}{2}\right)\sum_{m=1}^{d}(H_1)_m\otimes|k\rangle\langle k|,\sum_{m=1}^{d}(H_2)_m\otimes I^{\otimes n_p}\right]\right\|\\ \leq&\frac{1}{2}N_p\gamma_1\gamma_2 2n_x\sum_{m=1}^{d}\left(r-\frac{\sigma_m^2}{2}\right)\frac{\sigma_m^2\rho_{mm}}{2}.\end{aligned}$$

The error between $\mathbf{U}_*$ and $\mathbf{V}_{BS}$ is

$$\begin{aligned}&\|\mathbf{U}_*(\tau)-\mathbf{V}_{BS}(\tau)\|\\ \leq&\left\|\prod_{m=1}^{d}(U_2)_m-\tilde{\mathbf{V}}_2\right\|+\max_{0\leq k\leq N_p-1}\left|k-\frac{N_p}{2}\right|\left\|\prod_{m=1}^{d}(U_1)_m-\tilde{\mathbf{V}}_1\right\|\\ \leq&\frac{\gamma_2^2\tau^2 2n_x}{2}\sum_{m=1}^{d}\left(r-\frac{\sigma_m^2}{2}\right)^2+\frac{N_p}{2}\frac{\gamma_1^2\tau^2 2n_x}{2}\sum_{m=1}^{d}\left(\frac{\sigma_m^2\rho_{mm}}{2}\right)^2.\end{aligned}$$

To sum up, we have

$$\begin{aligned}\|\mathbf{U}_{BS}(\tau)-\mathbf{V}_{BS}(\tau)\|\leq&\|\mathbf{U}_{BS}(\tau)-\mathbf{U}_*(\tau)\|+\|\mathbf{U}_*(\tau)-\mathbf{V}_{BS}(\tau)\|\\ \leq&\frac{1}{4}\tau^2 n_x(N_p\gamma_1^2+2\gamma_2^2+2N_p\gamma_1\gamma_2)\sum_{m=1}^{d}c_m^2,\end{aligned}$$

where $c_m=\max_m\left(r-\frac{\sigma_m^2}{2},\frac{\sigma_m^2\rho_m}{2}\right)$. □

Lemma 5.5. *The approximated time evolution operator* $\mathbf{V}_{BS}(\tau)$ *in* (4.6) *can be implemented using* $\mathcal{Q}_{single} = O(dN_p n_x)$ *single-qubit gates and at most* $\mathcal{Q}_{V_{BS}} = O(dN_p n_x^2)$ CNOT *gates for* $n_x \geq 3$.

Proof. Based on the same discussion in the proof of Lemma 5.2, the single-qubit gates for $\mathbf{V}_1$ and $\mathbf{V}_2$ can be directly implemented for d dimension.

$$\begin{aligned}\mathcal{Q}_{single-d} =& \mathcal{Q}_{single-V_1} + \mathcal{Q}_{single-V_2} \\ =& d2^{n_p-1}(2 + 2n_x + 2n_x) + d(4n_x) = O(dN_p n_x).\end{aligned}$$

Similarly, for CNOT gate, we have

$$\mathcal{Q}_{\mathbf{BS}} = d\mathcal{Q}_{V_2} + d2^{n_p-1}\mathcal{Q}_{V_1} + d(2^{n_p} - 1)\mathcal{Q}_{c-V_1} = O(dN_p n_x^2).$$

□

Lemma 5.6. *Let* $\mathbf{H}_{BS}$ *be the Hamiltonian defined in Equation* (4.4)*. The time evolution operator* $\mathbf{U}_{BS}(T) = \exp(i\mathbf{H}_{BS}T)$ *for a time duration* T *can be implemented on a* $(dn_x + n_p)$*-qubit system using quantum circuits with* $O\left(\frac{1}{\varepsilon}dN_pT^2n_x^2(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)\sum_{m=1}^d c_m^2\right)$ *single-qubit gates and* $O\left(\frac{1}{\varepsilon}dN_pn_x^3T^2(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)\sum_{m=1}^d c_m^2\right)$ *CNOT gates, within an additive error of* ε.

Proof. To ensure that the error in the simulation over the total time T is bounded by ε, it suffices to divide the total time T into N intervals, where $N = T/\tau$, such that

$$\|\mathbf{U}_{BS}^N(\tau) - \mathbf{V}_{BS}^N(\tau)\| \leq N\|\mathbf{U}_{BS}(\tau) - \mathbf{V}_{BS}(\tau)\| \leq \varepsilon. \tag{5.4}$$

From Equation (5.3), we have

$$\frac{1}{4N}T^2n_x(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)\sum_{m=1}^d c_m^2 \leq \varepsilon,$$

which can be rearranged to yield

$$N \geq \frac{1}{4\varepsilon}T^2n_x(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)\sum_{m=1}^d c_m^2.$$

By Lemma 5.5, the unitary $\mathbf{V}_{BS}$ can be implemented using $O(dN_pn_x)$ single-qubit gates and at most $O(dN_pn_x^2)$ CNOT gates.

Thus, the implementation of $\mathbf{V}_{BS}^N$ requires $O\left(\frac{1}{\varepsilon}dN_pT^2n_x^2(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)\sum_{m=1}^d c_m^2\right)$ single-qubit gates and $O\left(\frac{1}{\varepsilon}dN_pn_x^3T^2(N_p\gamma_1^2 + 2\gamma_2^2 + 2N_p\gamma_1\gamma_2)\sum_{m=1}^d c_m^2\right)$ CNOT gates. □

Theorem 5.2. *For the d-dimensional Black-Scholes equation* (4.1)*, the state* $|\mathbf{U}(t)\rangle$*, where* $\mathbf{U}(\mathbf{t})$ *is the solution of Equation* (4.3) *with a mesh size* h*, can be prepared up to precision* ε' *using the Schrödingerisation method depicted in Figure 4.3 and Figure 3.5. This preparation can be achieved using at most* $\tilde{O}\left(\frac{dT^2\|\mathbf{U}(0)\|^3\sum_{m=1}^d c_m^2}{h^4\varepsilon^3\|\mathbf{U}(T)\|^3}\right)$ *single-qubit gates and* CNOT *gates.*

Proof. As discussed in Theorem 5.1, the computational complexity of the Schrödingerisation approach for the d-dimensional Black-Scholes equation with central difference discretization consists of three key components.

The complexity analysis of simulating the unitary $\mathbf{V}_{BS}^{N}(\tau)$ as concluded in Lemma 5.6 can be applied with $n_x = O(\log(L_x/h))$ and $\gamma_1 = \frac{1}{h^2 L_p}, \gamma_2 = \frac{1}{2h}$. This results in a computational cost of $O\left(\frac{dN_p^2 T^2 \log^3(L_x/h)\sum_{m=1}^{d} c_m^2}{h^4 L_p^2 \varepsilon'}\right)$, with $\varepsilon' = \|\mathbf{U}(T)\|\varepsilon/\|\mathbf{U}(0)\|$. Following Theorem 5.1, the required number of measurements and the error analysis of the output state preserve the same scaling. Then combining these components, we obtain the total complexity, $\tilde{O}\left(\frac{dT^2|\mathbf{U}(0)|^3\sum_{m=1}^{d} c_m^2}{h^4\varepsilon^3|\mathbf{U}(T)|^3}\right)$. □

Remark 5.1. *Introducing the inter-dimensional coupling (correlation coefficients $\rho_{mn} \neq 0, m \neq n$) requires additional quantum gates to implement the non-diagonal operations. If cross terms are considered, it is necessary to simulate the non-diagonal operator $\partial x_i \partial x_j$ which may introduce $O(d^2)$ terms (similar to the high-dimensional coupling in the heat equation [8]).*

Remark 5.2. *The Schrödingerisation method provides a promising pathway for quantum simulation of the Black-Scholes equation, particularly in high-dimensional settings such as multi-asset option pricing. Classical finite-difference methods suffer from the curse of dimensionality, requiring $(O(N_x^d))$ for d underlying assets. In contrast, the quantum approach scales as $\tilde{O}(d/\varepsilon^3)$, offering exponential memory savings and polynomial gate complexity in d. This advantage becomes pronounced for $d \geq 3$, where classical methods become computationally prohibitive.*

6 Numerical results

In this section, we use the proposed quantum circuit by the Qiskit package [7] and the classical difference method to verify the algorithm in the framework of the Black-Scholes model to ensure the feasibility of Schrödingerisation algorithm. Throughout the implementation phase, the *Statevector* simulator is utilized to retrieve the complete solution, while the *Estimator* function is employed for the measurement of observable.

Example 6.1. *The classic European call option price has an explicit expression at t time:*

$$u(x,t) = e^x N(d_1) - Ke^{-rt} N(d_2),$$

where

$$d_1 = \frac{x - \ln K + (r + \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}, \quad N(y) = \frac{1}{2\pi}\int_{-\infty}^{y} e^{-\frac{1}{2}s^2} ds.$$

To formulate the problem, parameters of equation are given: $T = 1$, rate $r = 0.02$, volatility $\sigma = 0.3$, exercise price $K = 30$, minimum asset price $s_{\min} = 10^{-4}$, maximum asset price $s_{\max} = 10K$, and the logarithm of $x \in L_x = [x_{\min}, x_{\max}] = [\ln(10^{-4}), \ln(10K)]$, auxiliary variable $p \in L_p = [p_{\min}, p_{\max}] = [-4\pi, 4\pi]$.

In this test, we choose two forms of initial values to simulate the problem. The first one is shown in (3.9) which is continuous but not in C^1. In order to increase the order of convergence of variables p, we consider a smoother initial solution discussed in [11],

$$g(p) = \begin{cases} (-3 + 3e^{-1})p^3 + (-5 + 4e^{-1})p^2 - p + 1, & p \in (-1, 0), \\ e^{-p}, \quad p \in (-\infty, -1] \cup [0, \infty), \end{cases}$$

and initial value of (3.8) is $\mathbf{v}(0) := \bar{\mathbf{u}}(0) \otimes g(p)$. The numerical results of two different initial solutions are denoted as u_h and u_h^*, respectively.

Denote the error as $\epsilon = |u - u_{numerical}|$, where $u_{numerical}$ denoted as u_h and u_h^*. In finance and real life applications, people focus on the function value evolution when S is in the neighborhood of the point $S = E$. Thus in the following representation of numerical results, we throw away the boring points and focus on the region that actually matters. In fact, we show the behavior of the function in the interval $S \in [0, 2E]$. We also show the error of the classical method and the Trotter splitting method when the same number of points are used to discretize the space dimension x. Our numerical experiments demonstrate that employing smoother initial conditions along the p-dimension yields solutions with significantly improved agreement to the analytical solution.

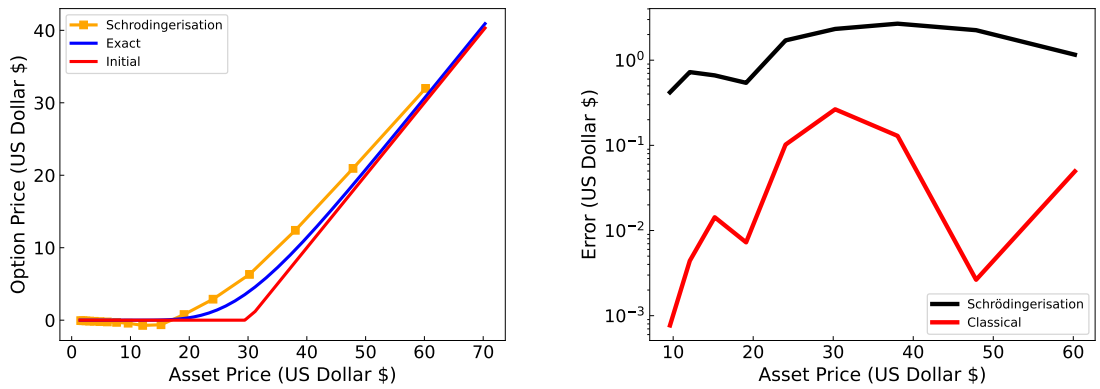


Fig. 6.1. Figures of the numerical solution u_h and error when $n_x = 6,\ n_p = 4$. CPU runtime: 00:20:17. Maximum RAM occupancy: 5.47 GB

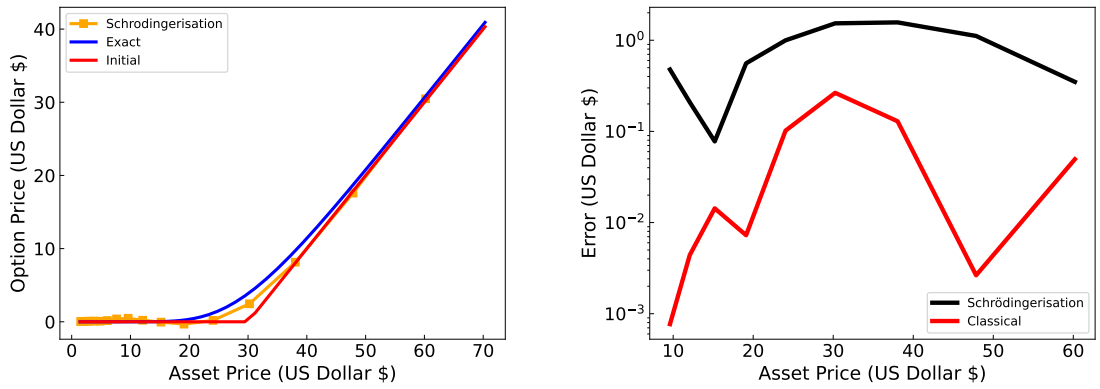


Fig. 6.2. Figures of the numerical solution u_h and error when $n_x = 6,\ n_p = 5$. CPU runtime: 01:11:15. Maximum RAM occupancy: 6.09 GB

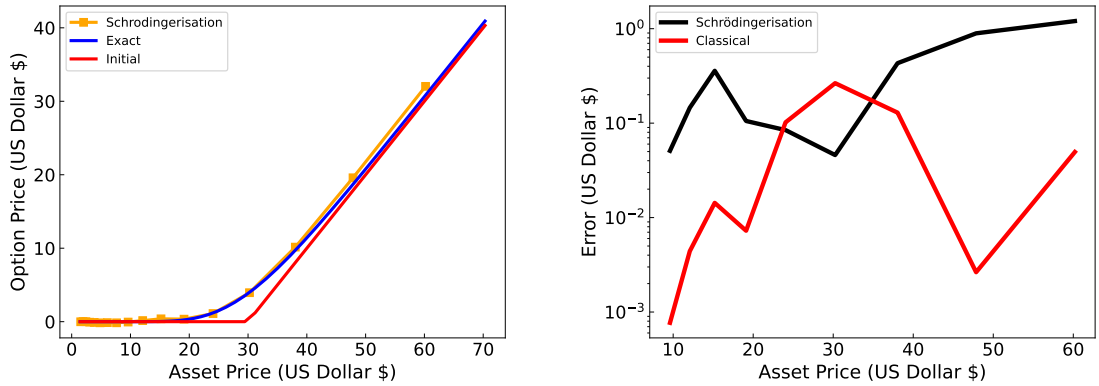


Fig. 6.3. Figures of the numerical solution u_h and error when $n_x = 6,\ n_p = 6$. CPU runtime: 03:37:11. Maximum RAM occupancy: 7.12 GB

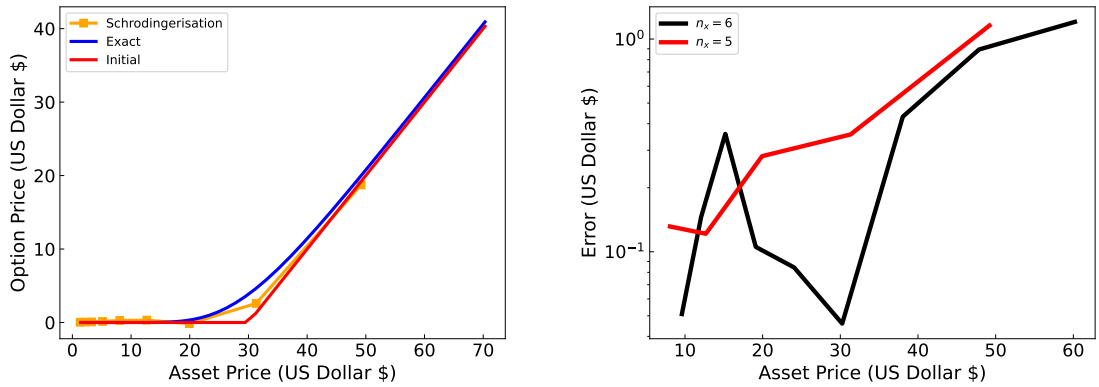


Fig. 6.4. Figures of the numerical solution u_h and error when $n_x = 5, 6$ and $n_p = 6$. CPU runtime: 00:18:08. Maximum RAM occupancy: 5.36 GB

To be clarified, the classical error in the figures comes from the directly solving of ODE (3.4) with matrix exponential. The point number used to discretize the space dimension in the original PDE is the sames as the number set in Schrödingerisation procedure. We use such classical error as a reference line and compare the right sub-figure in Figure 6.1, Figure 6.2 and Figure 6.3, one can tell that as the number of ancilla qubits n_p increases, the error produced by Schrödingerisation procedure steadily decrease. When $n_p = n_x = 6$, according to the analysis in [9], the relative error $\epsilon = \frac{\|w_h - w\|}{\|w\|}$ generated by the discretization on p dimension satisfies : $\epsilon \sim \Delta p = \pi/8$. w is the exact solution of ODE (3.4). Two pictures in Figure 6.3 verified the theory. Furthermore, comparing the left sub-figure in Figure 6.1, Figure 6.2 and Figure 6.3, one can also conclude that with the increase of n_p, numerical solution corresponding to the Schrödingerisation method converges to the exact solution. The convergence underscores the capability of the Schrödingerisation method to effectively approximate the solution. In real life applications, people are also concerned with where the inflection point of the function V with respect to S occurs at time t. For example, the asset price $S = E$, $E = 30$ in our example, is the inflection point when $t = 0$. Such point moves backward as t increase. One can tell from the figures that the Schrödingerisation method catches the inflection precisely. However due to the limitation of computing resources, the simulation with

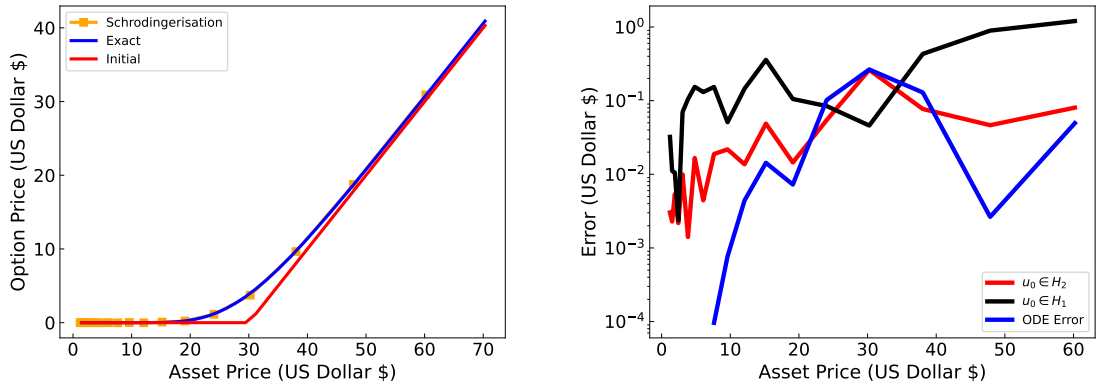


Fig. 6.5. Figures of the numerical solution u_h^* and error when $n_x = 6,\ n_p = 6$.

$(\Delta p, \Delta x)$	$(\frac{L_p}{2^7}, \frac{L_x}{2^6})$	order	$(\frac{L_p}{2^8}, \frac{L_x}{2^7})$	order	$(\frac{L_p}{2^9}, \frac{L_x}{2^8})$	order
$\|u_h - u\|_{L^2}$	6.698e-01	-	3.271e-01	1.03	1.776e-01	0.89
$\|u_h^* - u\|_{L^2}$	2.925e-01	-	7.540e-02	1.96	2.060e-02	1.88

TABLE 6.1. The convergence rates of two initial solution $\|u_h - u\|_{L^2}$ and $\|u_h^* - u\|_{L^2}$, using the classical finite difference method and the Schrödingerisation method.

parameters $n_x = 6,\ n_p = 6$ is the maximum computational load that the computer can handle. Therefore we do the simulation when $n_x = 5,\ n_p = 6$ to compare with the best cases. Figure 6.4 shows that when keeping n_p unchanged, more points on x dimension leads to better accuracy. Figure 6.5 shows that when we choose smoother initial condition $\mathbf{v}(0) := \bar{\mathbf{u}}(0) \otimes g(p)$, the error will further decrease.

In order to further illustrate the effectiveness of the Schrödingerisation algorithm and the effect of initial value, the convergence rates of two initial values with $\Delta t = 10^{-3}, T = 0.1$ are shown in Table 6. The results are obtained by using the central difference method for the Schrödingerisation method. According to the Table, second-order convergence rates of smooth initial value $\mathbf{v}(0) := \bar{\mathbf{u}}(0) \otimes g(p)$ can be obtained.

Example 6.2. *Consider the two-dimensional Black-Scholes pricing model for cash or valueless options*

$$\begin{cases} \frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma_1^2 \frac{\partial^2 u}{\partial x^2} + \frac{1}{2}\sigma_1^2 \frac{\partial^2 u}{\partial y^2} + \sigma_1\sigma_2\rho_{12}\frac{\partial^2 u}{\partial x \partial y} + \left(r - \frac{\sigma_1^2}{2}\right)\frac{\partial u}{\partial x} + \left(r - \frac{\sigma_2^2}{2}\right)\frac{\partial u}{\partial y} - ru, \\ u(x, y, 0) = u_0(x, y), \end{cases}$$

where $u_0(x, y) = c$ *with* $x > \ln(K_1), y > \ln(K_2)$*, otherwise* $u_0(x, y) = 0$.

Assume the amount paid by the right at maturity $c = 1$, exercise price $K_1 = K_2 = 50$, risk-free interest rate $r = 0.03$, asset volatility $\sigma_1 = \sigma_2 = 0.3$, asset correlation coefficient $\rho = 0.6$, the logarithm of asset price $x \in [x_{\min}, x_{\max}] = [\ln(10^{-6}), \ln(4K_1)]$, $y \in [y_{\min}, y_{\max}] = [\ln(10^{-6}), \ln(4K_2)]$, and auxiliary variable $p \in L_p = [-\pi L, \pi L]$, $L = 15$. Cash or valueless options satisfy the following boundary conditions

$$u(x_{\min}, y, \tau) = u(x, y_{\min}, \tau) = 0, \quad \forall x, y, \tau,$$
$$\frac{\partial u}{\partial x}(x_{\max}, y, \tau) = \frac{\partial u}{\partial y}(x_{\max}, y, \tau) = 0, \quad \forall x, y, \tau.$$

and the exact solution is

$$u(x, y, \tau) = ce^{-r\tau} B(d_x, d_y, \rho),$$

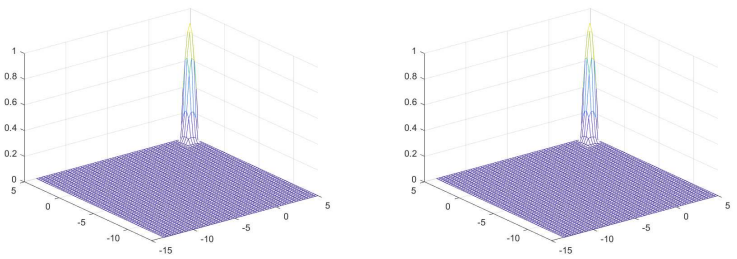
where

$$d_x = \frac{x - \ln K_1 + (r - \frac{1}{2}\sigma_1^2)\tau}{\sigma_1\sqrt{\tau}},$$

$$d_y = \frac{x - \ln K_2 + (r - \frac{1}{2}\sigma_2^2)\tau}{\sigma_2\sqrt{\tau}},$$

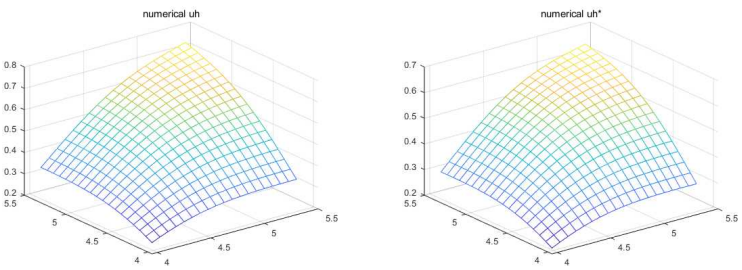
$$B(d_x, d_y, \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{d_x} \int_{-\infty}^{d_y} e^{-\frac{s_1^2 - 2\rho s_1 s_2 + s_2^2}{2(1-\rho^2)}} ds_2 ds_1.$$

The numerical results and the images in the upper right corner are shown in the Figure 6.6 and 6.7 using Matlab based on two initial values similar with the one-dimensional case. The results are obtained by using the central difference method for Schrödingerisation with the parameters $T = 1$, $\Delta t = 10^{-3}$, $n_x = 8$ and $n_p = 7$. The numerical results show that the Schrödingerisation method can simulate the price curve of dual-asset options well. In this test, the Schrödingerisation method can be directly applied to the Equation (4.3) without dilation. Any value of p that is slightly larger than $p^\diamond = \max(\lambda(\mathbf{A}_1)T, 0)$ can well retrieve the original solution.



(a) u_h (b) u_h^*

Fig. 6.6. The solutions of the 2d Black Sholes equation with two initial values .



(a) u_h (b) u_h^*

Fig. 6.7. The solutions of 2d Black Sholes equation (upper right).

7 Conclusion

In this paper, we develop a quantum computing framework for solving the Black-Scholes equation by combining the Trotter splitting method with the Schrödingerisation approach. We present

a practical implementation of quantum circuits based on operator splitting techniques, including detailed circuit designs for each computational component. Our analysis provides comprehensive quantum circuit descriptions and complexity estimates, demonstrating quantum advantages in high-dimensional settings compared to classical methods.

Numerical experiments performed on the Qiskit platform validate the convergence of our Schrödingerisation method for both one-dimensional and two-dimensional Black-Scholes equations, confirming its effectiveness for financial derivative pricing applications.

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